Description

Illustration





Figure caption

1. The top of the Picasso Tower, Madrid, built with lightweight concrete. 2. Bank of America Tower, Charlotte, NC. This 60-story building was constructed from lightweight concrete to minimize weight. Images used under license from Shutterstock.com

The material

Structural lightweight concrete comprises lightweight coarse aggregates (e.g. expanded shale, clay, air-cooled blast furnace slag or slate materials that have been fired in a rotary kiln). This reduces the strength-to-weight ratio of the concrete, and its dead-load which is important in structural applications such as bridge and long-span structural slab manufacture. When combined with steel reinforcing bars, the composite can be cast on site or delivered preformed and cured from a precasting plant, significantly improving its properties. Note that the properties quoted are for unreinforced concrete.

Compositional summary

Concrete: 0.6:1:0:4 Water:OPC:Fine:Coarse
Mix constituents and proportions are varied to produce widely ranging properties.

General properties

Density		1.4e3	-	2e3	kg/m^3
Price	*	0.0302	-	0.0453	GBP/kg

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✗
Services	

	✖			

Mechanical properties

Young's modulus	11	-	21	GPa
Shear modulus	* 4.6	-	8.9	GPa
Bulk modulus	* 5.8	-	11.1	GPa
Bending modulus	* 11	-	21	GPa
Poisson's ratio	0.17	-	0.2	
Yield strength (elastic limit)	* 1.1	-	2.8	MPa
Tensile strength	* 1.1	-	2.8	MPa
Compressive strength	* 11.3	-	28	MPa
Bending strength	3	-	17	MPa
Elongation	* 0.01	-	0.03	% strain
Hardness - Vickers	* 5.1	-	5.6	HV
Fatigue strength at 10^7 cycles	* 0.79	-	1.23	MPa
Fracture toughness	* 0.04	-	0.07	MPa.m^0.5
Mechanical loss coefficient (tan delta)	0.01	-	0.03	

Thermal and combustion properties

Thermal conductor or insulator?	Poor insulator			
Thermal resistivity	0.769	-	3.33	m.°C/W
Thermal expansion coefficient	7	-	12	µstrain/°C
Specific heat capacity	* 860	-	1.1e3	J/kg.°C
Melting point	930	-	1.2e3	°C
Maximum service temperature	560	-	640	°C
Minimum service temperature	-160	-	-150	°C
Flammability	Non-flammable			
Emissivity	0.63	-	0.97	

Hygro-thermal properties

Water vapor permeability	2e-12	-	6.01e-12	kg.m/s.m^2.Pa
Frost resistance	Average			

Electrical properties

Electrical conductor or insulator?	Good insulator			
Electrical resistivity	3.16e11	-	3.16e12	µohm.cm
Dielectric constant (relative permittivity)	* 8	-	12	
Dissipation factor (dielectric loss tangent)	* 0.001	-	0.01	
Dielectric strength (dielectric breakdown)	* 0.5	-	2	MV/m

Optical properties

Natural color

Gray

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Average
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Moldability	3	-	4	
Machinability	1			

Durability

Water (fresh)	Excellent
Water (salt)	Acceptable
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Unacceptable
Industrial atmosphere	Acceptable
Rural atmosphere	Acceptable
Marine atmosphere	Limited use

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.453	-	0.469	kg/kg
Embodied energy (virgin grade production)	* 4.49	-	4.97	MJ/kg
Water Usage	* 3.04	-	3.36	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 17	-	20.6	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.11	-	1.35	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.0453	-	0.0469	kg/kg
Embodied energy, recycling	* 0.629	-	0.695	MJ/kg
Recycle fraction in current supply	14			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Lightweight concrete, or more correctly called lightweight aggregate concrete, utilises artificial or natural lightweight aggregates in place of dense natural aggregates. Structural design methods are similar to those for dense concrete and dense reinforced concrete.

The benefits of using lightweight aggregate concrete include:
* Reduction in dead loads making savings in foundations and reinforcement.
* Improved thermal properties.
* Improved fire resistance.
* Savings in transporting and handling precast units on site.
* Reduction in formwork and propping.

The elastic modulus of lightweight concrete is lower than the equivalent strength dense concrete, which can affect deflections in reinforced structural elements (beams and slabs), but this is offset by the reduced self-weight.

The manufacture of lightweight aggregate concrete is more problematic than dense concrete, meaning it is more commonly used in precast beams, columns, walls and slabs (one-way, two-way, flat-plate, waffle slab and many others). Precast components are often used either as building enclosure panels or long-span post-tensioned structural sections.

Technical notes

Because lightweight aggregate has lower strength and higher porosity than either hardened cement paste or dense aggregate, the cement content has to be increased to obtain concrete of the same compressive strength; this can result in increased creep, shrinkage and a lower elastic modulus.

Standard design methods for reinforced structural lightweight concrete are similar to those for dense concrete, requiring only minor additional considerations.

As a result of increased porosity, which produces the lower density, lightweight aggregate has a high-water absorption, which can stiffen concrete as it is mixed. To offset this, the aggregate can be pre-coated with polymers to reduce absorption, but this can also make the aggregate float in the mix. It requires more care than for dense concrete mixes.

The quoted properties are for unreinforced lightweight concrete. However, in practice, they are significantly improved by reinforcement using steel bar or fibre.

Typical uses

Structural applications such as bridge and building manufacture.

Tradenames

Aggregate trade names include Leca. Lytag and Pellite as well as natural pumice.

Related standards

The properties of Cement and Concrete vary a great deal based on composition, age and processing. Standards exist to help ensure that consistent material properties can be achieved. As temperature ranges, humidity, raw materials and building practices vary from country to country so do standards. Here you can see a list of the main US, EU and UK standards related to this material. Other countries have their own standards, for example: Japan www.jsa.or.jp/en, China www.gbstandards.org, Australia www.standards.org.au.

ACI 301: Specifications for Structural Concrete
AS 3972: General purpose and blended cements
ASTM C150: Standard Specification for Portland Cement
ASTM C1697: Standard Specification for Blended Supplementary Cementitious Materials
ASTM C55: Standard Specification for Concrete Building Brick
ASTM C90: Standard Specification for Loadbearing Concrete Masonry Units
EN 13670: Execution of concrete structures
EN 197-1: Cement. Composition, specifications and conformity criteria for common cements
EN 1992-1-1: General rules, and rules for buildings
EN 1992-1-4: Lightweight aggregate concrete with closed structure
EN 1992-1-6: Plain concrete structures
EN 206: Concrete. Specification, performance, production and conformity
BS 8500: Concrete. Complementary British Standard to BS EN 206
CSI/CSC 03 00 00: Concrete
CSI/CSC 03 11 00: Concrete Forming
CSI/CSC 03 30 00: Cast-in-place Concrete
CSI/CSC 03 31 00: Structural Concrete
CSI/CSC 03 33 16: Lightweight Architectural Concrete
CSI/CSC 03 40 00: Precast Concrete
CSI/CSC 04 20 00: Unit Masonry
CSI/CSC 04 22 00: Concrete Unit Masonry
CSI/CSC 04 23 00: Glass Unit Masonry

Further reading

Kind-Barkauskaus, F. Kauhsen, B. Polonyi, S. Brandt, J. "Concrete Construction Manual" Birkhauser-Publishers for Architecture, Switzerland (2002)
Nilson, A.H. and G. Winter, "Design of Concrete Structures", McGraw-Hill Book Company, New York, USA (1986)
MacGregor, J.G. "Reinforced Concrete, Mechanics and Design, 3rd edition", Prentice Hall, USA (1997)
Wilson, A. (1993) Cement and Concrete: Environmental Considerations. Environmental Building News, Vol. 2, No. 2: (www.buildinggreen.com).
van Oss, H.G. Padovani, A.C. (2002) Cement Manufacture and the Environment, Part 1: Chemistry and Technology. Journal of Industrial Ecology, Vol. 6, No. 1, pp.89-105.
Bentur, A., (2002) Cementitious Materials - Nine Millennia and a New Century: Past, Present, and Future. American Society of Civil Engineers, Journal of Materials in Civil Engineering, Vol.14, No. 1: pp. 2-22.

Links

Producers
Reference

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.